

Introduction

Housing prices have become an important issue in many cities around the world.

Whether in Shanghai, Toronto, or many other places, rising housing prices affect home buyers, sellers, and policymakers. Understanding which housing characteristics are associated with housing prices can help people better understand the housing market and support more informed housing decisions. In this project, the Ames Housing dataset is used as a case study to examine how different housing characteristics are associated with residential sale prices.

The main research question is: Which housing characteristics are most strongly associated with residential housing prices? Based on previous studies, we expect that larger houses, newer houses, houses with higher overall quality, and houses located in more desirable neighborhoods are associated with higher sale prices.

The dataset used in this project comes from the AmesHousing package in R and is based on the Ames, Iowa housing dataset originally introduced by De Cock (2011).

The original data were collected by the Ames Assessor's Office and contain information on 2,930 residential property sales. For this analysis, sale price is used as the response variable, while above-ground living area, basement area, overall quality, year built, neighborhood, and garage capacity are included as explanatory variables.

Previous studies show that housing prices are influenced by both structural characteristics and location. Sirmans et al. (2005) reviewed many studies on housing prices and concluded that house size, housing quality, garage space, and neighborhood

are among the most important factors associated with housing values. Similarly, Zhang (2021) used multiple linear regression to study residential housing prices and found that house size, quality, and location explained a large proportion of the variation in housing prices. These findings suggest that both physical characteristics and neighborhood should be considered when studying housing values.

Multiple linear regression is an appropriate method for this study because housing prices are affected by several housing characteristics at the same time. It allows us to examine the relationship between each housing characteristic and sale price while controlling for the effects of the other variables. This approach makes it possible to compare the relative importance of different housing characteristics. The purpose of this project is to better understand these relationships, rather than simply build the most accurate prediction model.

Data Description

Table 1. Number of Missing Values in Each Variable

Number of Missing Values in Each Variable						
sale_price	living_area	basement_area	overall_quality	year_built	neighborhood	garage_capacity
0	0	0	0	0	0	0

The data used in this project come from the AmesHousing package in R, which is based on the Ames Housing dataset originally introduced by De Cock (2011). The original data were collected by the Ames Assessor's Office and include information on 2,930 residential property sales in Ames, Iowa. The dataset contains detailed information on both housing characteristics and sale prices, making it suitable for studying the factors associated with residential housing values.

For this analysis, sale price is selected as the response variable. Six housing characteristics are included as explanatory variables: above-ground living area, basement area, overall quality, year built, neighborhood, and garage capacity. These variables describe different aspects of a house, including its size, age, construction quality, parking capacity, and location

Table 2. Summary Statistics for Numerical Predictors

Summary Statistics for Numerical Predictors				
Variable	Mean	SD	Min	Max
Living Area	1499.69	505.51	334	5642
Basement Area	1051.26	440.97	0	6110
Year Built	1971.36	30.25	1872	2010
Garage Capacity	1.77	0.76	0	5

As shown in Table 1, no missing values were found in any of the selected variables. Therefore, all 2,930 observations were retained for the analysis, and no additional data cleaning was required. Table 2 summarizes the four numerical predictors used in the analysis. Living area and basement area show considerable variation across houses, while year built ranges from 1872 to 2010. Garage capacity varies from 0 to 5 spaces, indicating differences in parking availability among properties.

Exploratory Data Analysis

Exploratory data analysis was conducted to better understand the distribution of housing sale prices and the relationships between sale price and the selected housing characteristics.

Table3. Summary Table for Sale Price

Summary Statistics for Housing Sale Price	
Statistic	Value
Minimum	12789.00
First Quartile	129500.00
Median	160000.00
Mean	180796.06
Third Quartile	213500.00
Maximum	755000.00
Standard Deviation	79886.69
Interquartile Range	84000.00

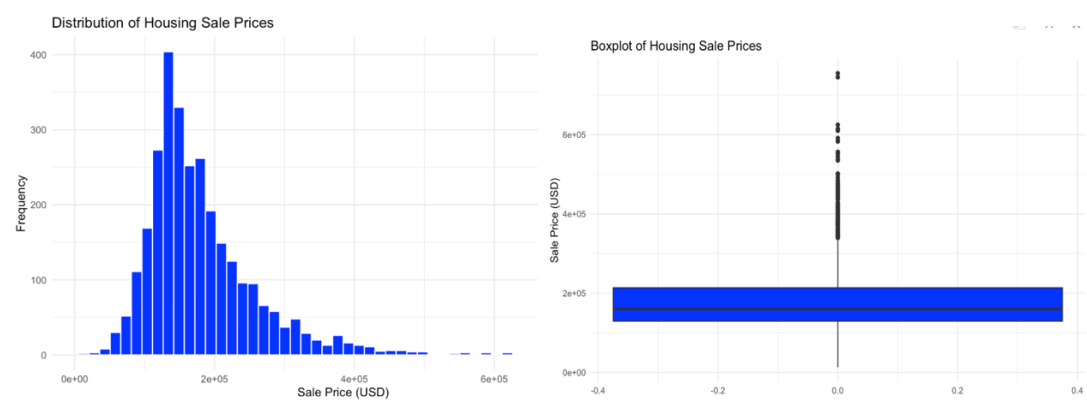


Figure 1 (left) . Distribution of Housing Sale Price

Figure 2 (Right) . Boxplot of Housing Sale Price

As shown in Table 3, housing sale prices vary substantially across the dataset. The large difference between the minimum and maximum sale prices, together with the relatively large standard deviation, indicates considerable variation in housing values. Figure 1 shows that the distribution of sale prices is positively skewed. Most houses are sold at relatively moderate prices, while a smaller number of expensive houses create a long right tail. Figure 2 also identifies several observations with unusually high sale prices. These observations are reasonable for housing data and are retained in the analysis. However, the skewed distribution suggests that the response variable may not satisfy the normality assumption and that a transformation of the response variable may be considered during model development.

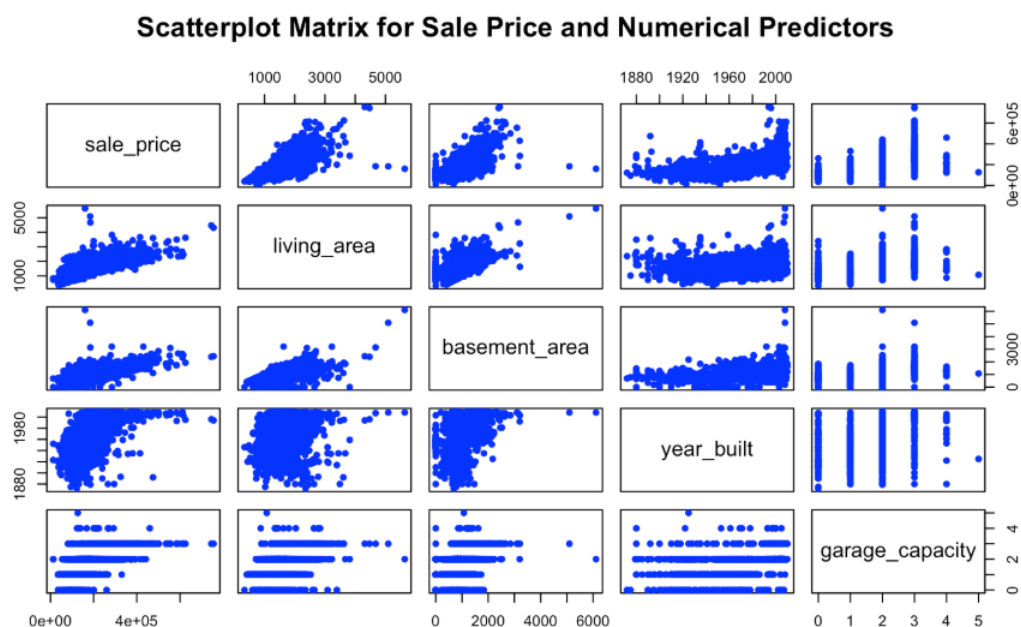


Figure 3. Scatterplot Matrix Between Sale Price and Numerical Predictors

Figure 3 provides an overall view of the relationships among sale price and the four numerical predictors. The scatterplot matrix suggests that sale price is positively associated with all four numerical variables. It also provides a preliminary view of the relationships among the predictors themselves, which is useful before fitting the multiple linear regression model.

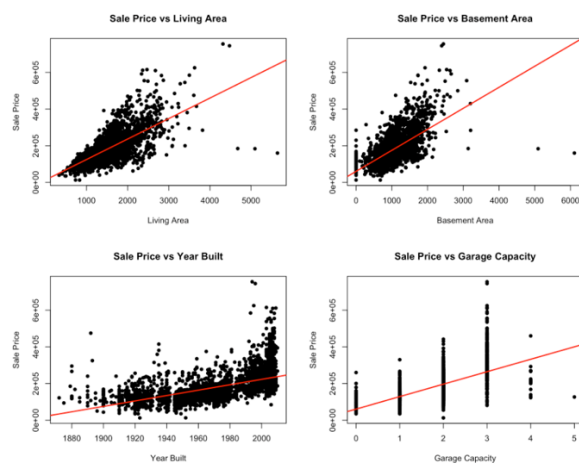


Figure 4. Relationships Between Housing Sale Price and Numerical Predictors

To better examine each numerical predictor individually, Figure 4 presents scatterplots of sale price against living area, basement area, year built, and garage capacity. Living area appears to have the strongest positive association with sale price, suggesting that larger houses generally sell for higher prices. Basement area also shows a positive relationship, although the association appears weaker than that of above-ground living area. Houses built more recently also tend to have higher sale prices. Garage capacity shows an overall positive association with sale price, indicating that houses with more parking spaces generally have higher sale prices.

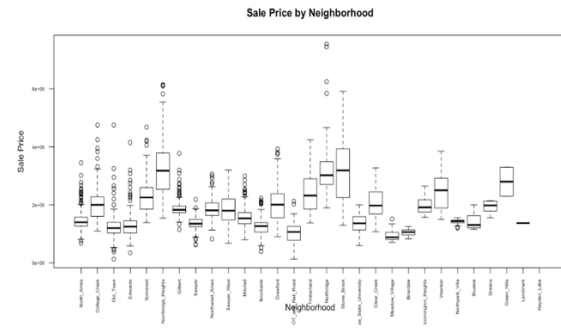


Figure 5 (Left) . Sale Price by Overall Quality Boxplot

Figure 6 (Right) . Sale Price by Neighborhood Boxplot

The relationships between the categorical predictors and sale price are shown in Figure 5 and Figure 6. Figure 5 shows a clear upward trend in sale price as overall quality increases. Houses with higher quality ratings generally have substantially higher median sale prices than houses with lower quality ratings, suggesting that construction quality is strongly associated with housing value.

Figure 6 illustrates the variation in sale prices across neighborhoods. Some neighborhoods have noticeably higher median sale prices than others, while several neighborhoods show relatively lower housing values. These differences suggest that neighborhood location may also be an important factor associated with residential housing prices.

Overall, the exploratory data analysis suggests that both structural characteristics and neighborhood location are associated with housing sale prices. The observed patterns support the use of multiple linear regression to further examine these relationships. In addition, the right skewed distribution of the response variable indicates that a

transformation may improve the regression model and will be investigated in the following section.

Regression Model

$$SalePrice_i = \beta_0 + \beta_1 GrLivArea_i + \beta_2 TotalBsmstSF_i + \beta_3 YearBuilt_i + \beta_4 GarageArea_i + \alpha_{q(i)} + \delta_{n(i)} + \epsilon_i,$$

Table 4. Coefficient Estimates for the Initial Regression Model

Coefficient Estimates for the Initial Regression Model						
Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-617990.5316	86583.9375	-7.1375	0.0000	-787763.0574	-448218.0057
living_area	46.1330	1.6134	28.5942	0.0000	42.9696	49.2965
basement_area	19.8542	1.7766	11.1757	0.0000	16.3708	23.3377
overall_qualityPoor	16400.6276	17973.0051	0.9125	0.3616	-18840.5795	51641.8348
overall_qualityFair	22945.6364	16470.7621	1.3931	0.1637	-9349.9945	55241.2674
overall_qualityBelow_Average	31178.9733	15879.9720	1.9634	0.0497	41.7552	62316.1915
overall_qualityAverage	40296.2928	15803.4011	2.5498	0.0108	9309.2137	71283.3719
overall_qualityAbove_Average	48473.8315	15840.9705	3.0600	0.0022	17413.0869	79534.5761
overall_qualityGood	61561.3754	15925.5783	3.8656	0.0001	30334.7331	92788.0177
overall_qualityVery_Good	90567.2102	16096.5012	5.6265	0.0000	59005.4246	122128.9958
overall_qualityExcellent	157750.1685	16461.5892	9.5829	0.0000	125472.5237	190027.8132
overall_qualityVery_Excellent	199616.5457	17331.0032	11.4602	0.0000	164634.1666	232598.9248
year_built	318.2088	43.5881	7.3004	0.0000	232.7419	403.6757
neighborhoodCollege_Creek	7063.1246	2977.5143	2.3722	0.0177	1224.8578	12901.3913
neighborhoodOld_Town	-9917.9897	3014.5461	-3.2900	0.0010	-15828.8678	-4007.1116
neighborhoodEdwards	-8730.0150	2771.5714	-3.1498	0.0017	-14164.4718	-3295.5581
neighborhoodSomerset	9926.3921	3520.2998	2.8198	0.0048	3023.8394	16828.9448
neighborhoodNorthridge_Heights	34908.3923	3908.9283	8.9304	0.0000	27243.8226	42572.9621
neighborhoodGilbert	1842.5581	3423.6368	0.5382	0.5905	-4870.4591	8555.5753
neighborhoodSawyer	57.5733	2978.8977	0.0193	0.9846	-5783.4061	5898.5526
neighborhoodNorthwest_Ames	1236.7492	3293.6375	0.3755	0.7073	-5221.3673	7694.8657
neighborhoodSawyer_West	77.7257	3497.3991	0.0222	0.9823	-6779.9236	6935.3750
neighborhoodMitchell	1019.6330	3423.0377	0.2979	0.7658	-5692.2096	7731.4755
neighborhoodBrookside	1765.2572	3580.4236	0.4930	0.6220	-5255.1854	8785.6998
neighborhoodCrawford	32831.5386	3591.1986	9.1422	0.0000	25789.9687	39873.1085
neighborhoodIowa_DOT_and_Rail_Road	-10030.9081	3899.2152	-2.5725	0.0101	-17676.4326	-2385.3836
neighborhoodTimberland	20195.3883	4370.6297	4.6207	0.0000	11625.5212	28765.2555
neighborhoodNorthridge	47330.7180	4728.0016	10.0107	0.0000	38060.1211	56601.3149
neighborhoodStone_Brook	43776.9031	5316.9042	8.2335	0.0000	33351.5946	54202.2115
neighborhoodSouth_and_West_of_Iowa_State_University	-8137.2631	4986.4015	-1.6319	0.1028	-17914.5266	1640.0004
neighborhoodClear_Creek	27484.1440	5015.2817	5.4801	0.0000	17650.2525	37318.0354
neighborhoodMeadow_Village	-18048.1390	5660.7771	-3.1883	0.0014	-29147.7085	-6948.5694
neighborhoodBriardale	-24772.5264	6047.6098	-4.0963	0.0000	-36630.5917	-12914.4610
neighborhoodBloomington_Heights	-4395.6536	6513.4895	-0.6749	0.4998	-17167.2091	8375.9019
neighborhoodVeenker	23680.2391	6744.3427	3.5111	0.0005	10456.0300	36904.4481
neighborhoodNorthpark_Villa	-17118.7872	6802.3281	-2.5166	0.0119	-30456.6933	-3780.8811
neighborhoodBlueste	-19204.1970	10154.9528	-1.8911	0.0587	-39115.8807	707.4868
neighborhoodGreens	-5681.6403	11491.1531	-0.4944	0.6210	-28213.3263	16850.0456
neighborhoodGreen_Hills	112550.9511	22316.2048	5.0435	0.0000	68793.6612	156308.2410
neighborhoodLandmark	-22234.6923	31414.4616	-0.7078	0.4791	-83831.7118	39362.3272
garage_capacity	10578.7813	1078.1690	9.8118	0.0000	8464.7232	12692.8395

To investigate the relationships between housing characteristics and housing sale prices, an initial multiple linear regression model was fitted using the original sale price as the response variable. The model included living area, basement area, overall quality, year built, neighborhood, and garage capacity as explanatory variables. These variables were selected based on previous literature and the exploratory data analysis

presented in the previous sections.

The estimated regression coefficients are presented in Table 4. Most of the continuous predictors, including living area, basement area, year built, and garage capacity, have positive estimated coefficients and are statistically significant at the 5% significance level. Several levels of overall quality and neighborhood are also statistically significant, indicating that these housing characteristics are associated with housing sale prices. However, some categories, such as Poor and Fair overall quality, together with several neighborhood categories including Gilbert, Sawyer, and Northwest Ames, are not statistically significant. This suggests that not every category differs significantly from the reference group in the initial model.

Although many predictors appear to be statistically significant, statistical significance alone is not sufficient to conclude that the model is appropriate. The validity of a multiple linear regression model also depends on whether the underlying assumptions are satisfied. Therefore, diagnostic plots were examined before interpreting the regression coefficients in detail.

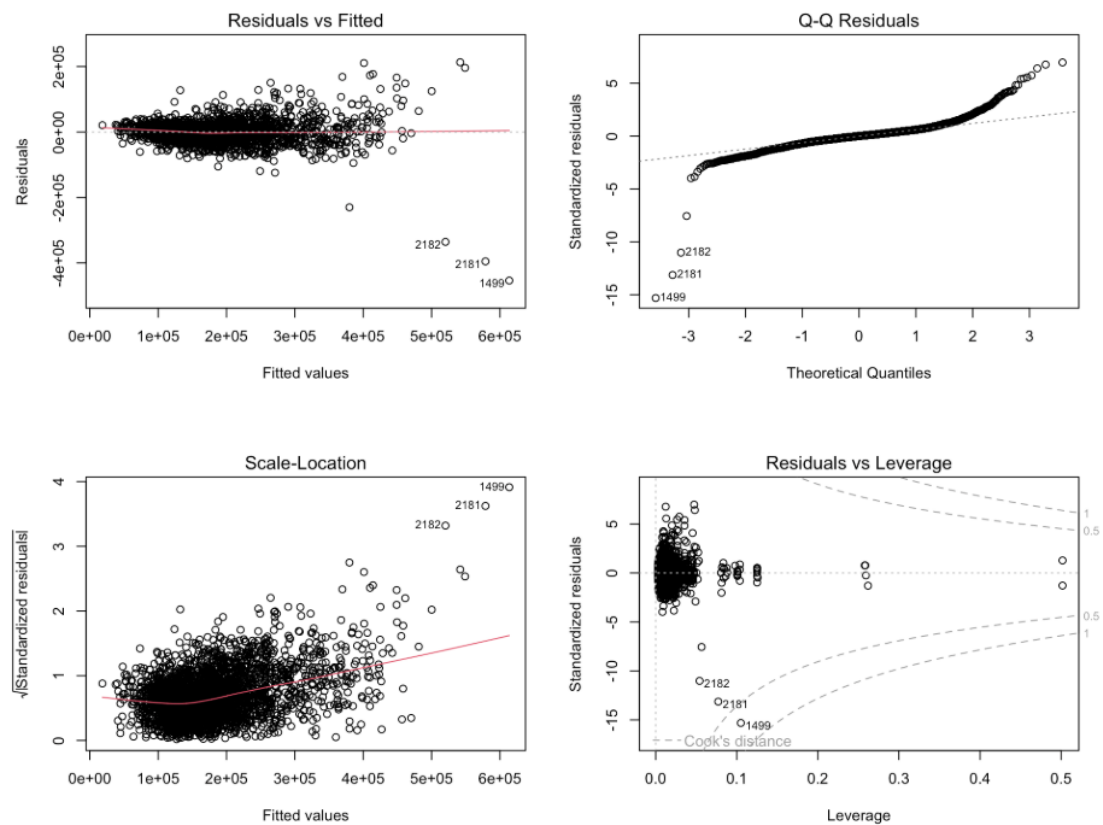


Figure 7. Standard Diagnostic Plots for the Initial Multiple Linear Regression Model

The standard diagnostic plots for the initial multiple linear regression model are presented in Figure 7. The Residuals versus Fitted plot shows that the spread of the residuals becomes larger as the fitted values increase, suggesting that the constant variance assumption may not be satisfied. In addition, several observations with relatively large residuals are visible at higher fitted values.

The Normal Q-Q plot shows clear departures from the reference line, particularly in both tails of the distribution. This indicates that the residuals are not normally distributed and that the normality assumption is violated.

The Scale-Location plot also displays an increasing trend in the spread of the standardized residuals as the fitted values increase. This provides further evidence of heteroscedasticity in the initial model.

Finally, the Residuals versus Leverage plot identifies a few observations with relatively high leverage. However, most observations remain well within the Cook's distance reference curves, suggesting that no extremely influential observations dominate the fitted model.

Overall, the diagnostic plots indicate that the assumptions of normality and constant variance are not fully satisfied for the initial regression model. Therefore, additional model refinement is required before interpreting the regression coefficients. In the following section, a Box-Cox transformation is considered to improve the model assumptions.

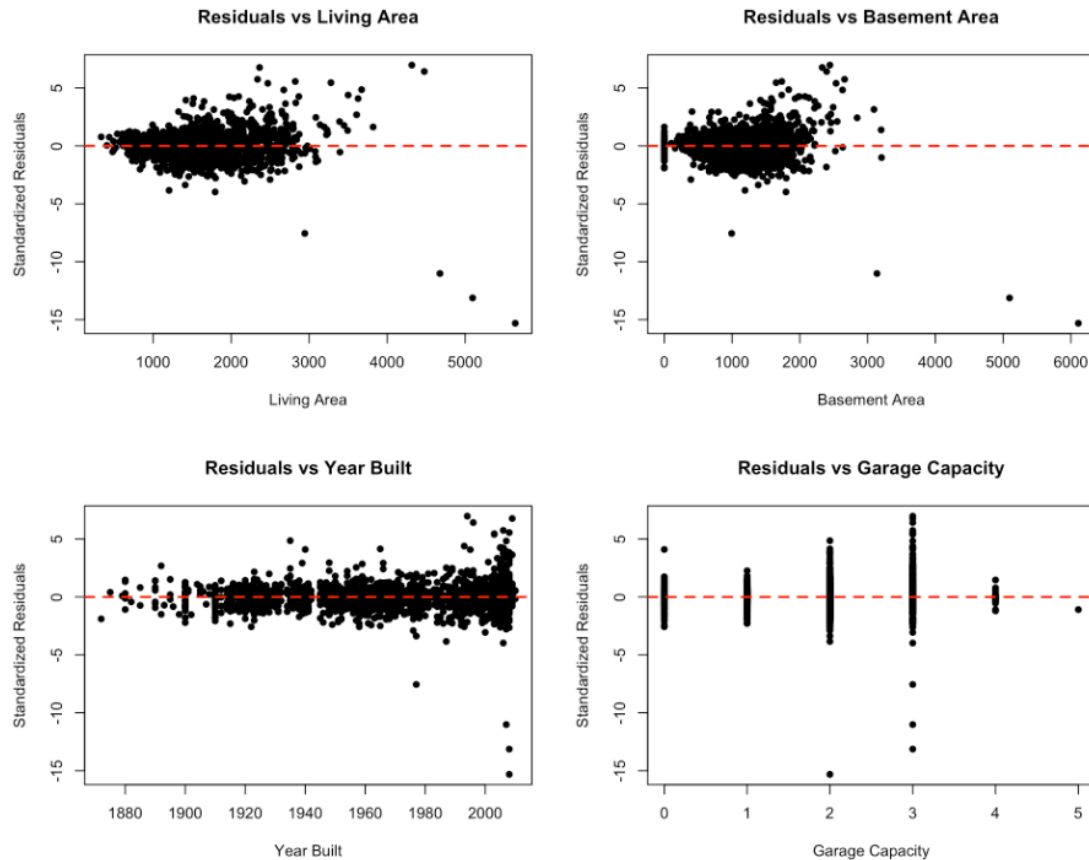


Figure 8. Residual Plots for the Numerical Predictors

The standardized residual plots for the numerical predictors are presented in Figure 8.

These plots are used to examine whether the residuals are randomly distributed around zero and whether the assumption of constant variance is reasonable for each numerical predictor.

For living area, the residuals are generally centered around zero; however, the spread of the residuals becomes noticeably larger as living area increases. Houses with relatively large living areas also contain several observations with unusually large positive and negative residuals. This pattern suggests that the variability of the residuals increases with house size and provides evidence that the constant variance assumption may not be fully satisfied.

A similar pattern is observed for basement area. Most observations are concentrated at smaller basement sizes, while houses with larger basements display greater variability in the residuals. Several extreme observations with large negative residuals are also present, indicating that the model tends to predict some large-basement houses less accurately than others.

For year built, the residuals remain approximately centered around zero across most construction years. However, the spread of the residuals gradually increases for newer houses, particularly for properties built after approximately 1990. This suggests that the variability of sale prices is larger among newer houses, which again indicates possible heteroscedasticity.

The residual plot for garage capacity shows that the residuals are generally centered around zero for each garage size category. Nevertheless, the variability differs across garage capacities, with houses having two or three garage spaces displaying noticeably larger spreads than houses with no or one garage space. A few observations with unusually large residuals are also present in these categories.

Overall, although the residuals are generally centered around zero for all four numerical predictors, the increasing spread observed in several plots suggests that the assumption of constant variance is not fully satisfied. These results are consistent with the conclusions obtained from the standard diagnostic plots in Figure 7.

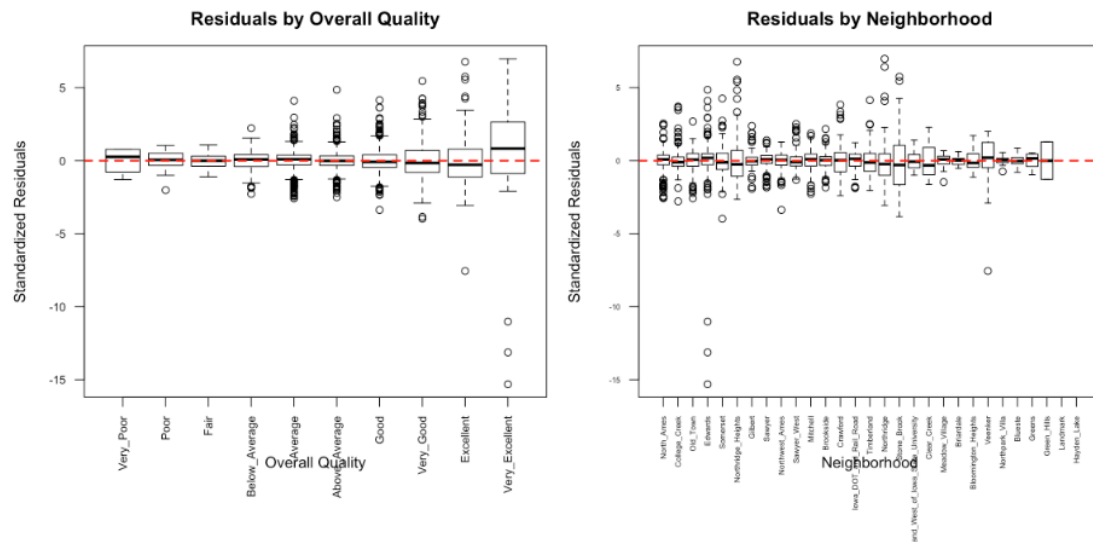


Figure 9. Residual Plots for the Categorical Predictors

Figure 9 presents the standardized residuals for the two categorical predictors included in the initial regression model. Boxplots are used to examine whether the residuals remain centered around zero within each category and whether the variability of the residuals differs across groups.

For overall quality, the median residual for nearly every quality level is close to zero, indicating that the model does not systematically overestimate or underestimate sale prices within most quality categories. However, the spread of the residuals increases for houses with higher quality ratings, particularly for the Excellent and Very Excellent categories. These groups also contain several extreme observations, suggesting that high-quality houses exhibit greater variation in sale prices than lower-quality houses.

For neighborhood, the residuals are also generally centered around zero across most neighborhoods, indicating that the neighborhood effects included in the model

adequately account for average differences among locations. Nevertheless, the variability of the residuals differs across neighborhoods, and several neighborhoods contain unusually large positive or negative residuals. These observations suggest that additional housing characteristics not included in the current model may still influence sale prices within certain neighborhoods.

Overall, the categorical residual plots indicate that the model captures the general effects of overall quality and neighborhood reasonably well. However, the unequal spread of the residuals across several categories provides further evidence that the assumption of constant variance is not completely satisfied.

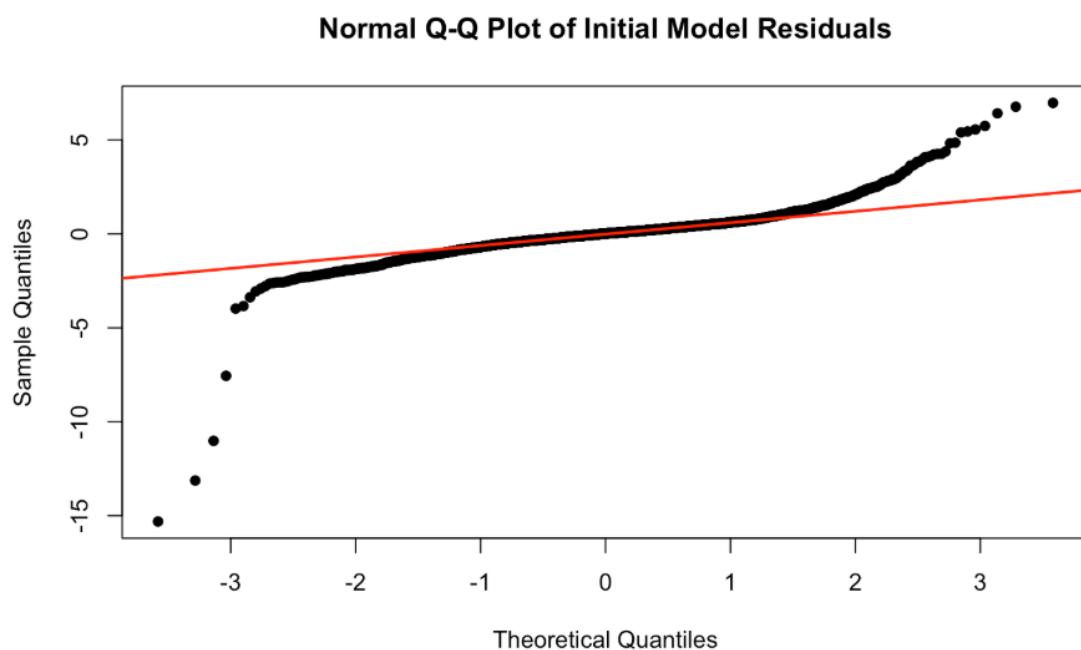


Figure 10. Normal Q-Q Plot of Standardized Residuals

The Normal Q-Q plot shown in Figure 10 provides a more detailed assessment of the normality assumption for the standardized residuals. If the residuals followed a normal distribution, the observations would lie approximately along the reference line

throughout the plot.

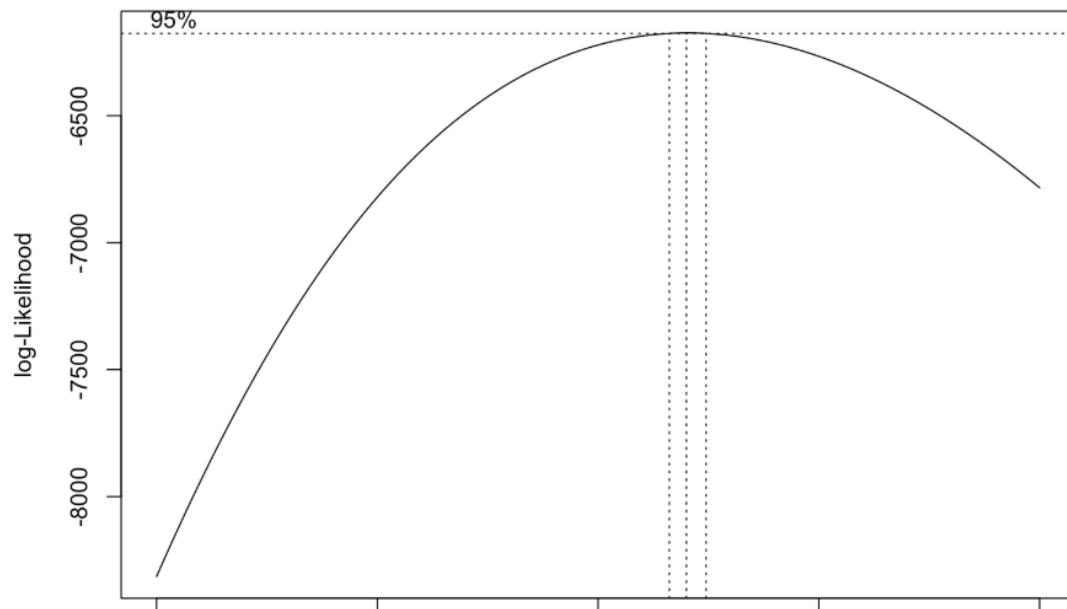
In this case, the residuals follow the reference line reasonably well near the center of the distribution. However, substantial departures from the line are observed in both the lower and upper tails. The lower tail falls well below the reference line, while the upper tail curves noticeably above it. These departures indicate that the residual distribution has heavier tails than expected under normality.

The presence of several extreme observations further contributes to the departure from normality. Because the normality assumption is important for statistical inference in multiple linear regression, the observed deviations suggest that the response variable may require transformation before the final model is selected.

Taken together with the previous diagnostic plots, the Normal Q-Q plot indicates that the initial regression model does not fully satisfy the assumptions of normality and constant variance. Therefore, model refinement is necessary. In the following section, a Box-Cox transformation is applied to improve the distribution of the residuals and the overall model performance.

Transformed Model (Log- Likelihood)

Figure 11. Box-Cox Profile Log-Likelihood



The diagnostic results from the initial regression model suggested that the assumptions of normality and constant variance were not fully satisfied. To improve these model assumptions, a Box-Cox transformation was considered for the response variable.

Figure 11 presents the profile log-likelihood for different values of the Box-Cox transformation parameter, λ . The curve reaches its maximum at approximately $\lambda = 0.25$, indicating that a power transformation is more appropriate than using the original response variable. The estimated value also falls within the 95% confidence interval shown in the figure, supporting the choice of $\lambda = 0.25$ for the subsequent analysis.

The purpose of the Box-Cox transformation is to stabilize the variance of the response

variable and improve the normality of the residuals rather than to change the relationships between the predictors and housing sale price. Therefore, the transformed response variable was used to fit the regression model in the following analysis, and the resulting diagnostic plots were compared with those from the initial model.

Table 5. Coefficient Estimates for the Box-Cox Transformed Regression Model

Coefficient Estimates for the Box-Cox Transformed Model						
Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-31.0532	8.5501	-3.6319	0.0003	-47.8181	-14.2882
living_area	0.0049	0.0002	30.8271	0.0000	0.0046	0.0052
basement_area	0.0021	0.0002	11.8670	0.0000	0.0017	0.0024
overall_qualityPoor	3.8377	1.7748	2.1623	0.0307	0.3576	7.3177
overall_qualityFair	8.8908	1.6265	5.4663	0.0000	5.7016	12.0799
overall_qualityBelow_Average	11.4622	1.5681	7.3094	0.0000	8.3874	14.5369
overall_qualityAverage	13.6029	1.5606	8.7166	0.0000	10.5429	16.6628
overall_qualityAbove_Average	14.9645	1.5643	9.5663	0.0000	11.8973	18.0317
overall_qualityGood	16.4420	1.5726	10.4550	0.0000	13.3584	19.5256
overall_qualityVery_Good	18.8435	1.5895	11.8548	0.0000	15.7268	21.9602
overall_qualityExcellent	23.1071	1.6256	14.2147	0.0000	19.9197	26.2945
overall_qualityVery_Excellent	23.3101	1.7114	13.6203	0.0000	19.9544	26.6659
year_built	0.0410	0.0043	9.5303	0.0000	0.0326	0.0495
neighborhoodCollege_Creek	0.6965	0.2940	2.3689	0.0179	0.1200	1.2730
neighborhoodOld_Town	-1.5245	0.2977	-5.1212	0.0000	-2.1082	-0.9408
neighborhoodEdwards	-1.2796	0.2737	-4.6754	0.0000	-1.8163	-0.7430
neighborhoodSomerset	1.0559	0.3476	3.0375	0.0024	0.3743	1.7375
neighborhoodNorthridge_Heights	2.7815	0.3860	7.2059	0.0000	2.0246	3.5384
neighborhoodGilbert	0.2219	0.3381	0.6564	0.5116	-0.4410	0.8848
neighborhoodSawyer	-0.0494	0.2942	-0.1681	0.8666	-0.6262	0.5274
neighborhoodNorthwest_Ames	0.2183	0.3252	0.6711	0.5022	-0.4195	0.8560
neighborhoodSawyer_West	-0.2275	0.3454	-0.6589	0.5100	-0.9047	0.4496
neighborhoodMitchell	0.1702	0.3380	0.5035	0.6147	-0.4926	0.8330
neighborhoodBrookside	-0.2512	0.3536	-0.7105	0.4775	-0.9445	0.4421
neighborhoodCrawford	3.5943	0.3546	10.1353	0.0000	2.8989	4.2896
neighborhoodIowa_DOT_and_Rail_Road	-2.4378	0.3850	-6.3312	0.0000	-3.1928	-1.6828
neighborhoodTimberland	1.9831	0.4316	4.5947	0.0000	1.1368	2.8293
neighborhoodNorthridge	3.0425	0.4669	6.5166	0.0000	2.1271	3.9580
neighborhoodStone_Brook	3.1615	0.5250	6.0214	0.0000	2.1320	4.1910
neighborhoodSouth_and_West_of_Iowa_State_University	-0.9011	0.4924	-1.8299	0.0674	-1.8666	0.0644
neighborhoodClear_Creek	3.1197	0.4953	6.2992	0.0000	2.1486	4.0908
neighborhoodMeadow_Village	-3.4899	0.5590	-6.2431	0.0000	-4.5860	-2.3938
neighborhoodBriardale	-4.5144	0.5972	-7.5592	0.0000	-5.6853	-3.3434
neighborhoodBloomington_Heights	-0.4981	0.6432	-0.7744	0.4388	-1.7593	0.7631
neighborhoodVeenker	2.4052	0.6660	3.6114	0.0003	1.0993	3.7111
neighborhoodNorthpark_Villa	-2.2948	0.6717	-3.4163	0.0006	-3.6119	-0.9777
neighborhoodBlueste	-2.5451	1.0028	-2.5380	0.0112	-4.5114	-0.5789
neighborhoodGreens	0.2457	1.1347	0.2165	0.8286	-1.9793	2.4707
neighborhoodGreen_Hills	10.9210	2.2037	4.9557	0.0000	6.6000	15.2421
neighborhoodLandmark	-3.0616	3.1022	-0.9869	0.3238	-9.1443	3.0211
garage_capacity	1.2777	0.1065	12.0009	0.0000	1.0690	1.4865

After applying the Box-Cox transformation, a new multiple linear regression model was fitted using the transformed response variable while keeping the same set of

explanatory variables. This allows the effect of the transformation to be evaluated without changing the model structure.

The estimated regression coefficients for the transformed model are presented in Table 5. Compared with the initial regression model, most of the continuous predictors, including living area, basement area, year built, and garage capacity, remain statistically significant at the 5% significance level. Similarly, many levels of overall quality and neighborhood continue to show statistically significant associations with the transformed response variable. Only a small number of neighborhood categories remain statistically insignificant, indicating that the overall conclusions from the initial model are largely preserved after the transformation.

Although the numerical values of the regression coefficients differ from those in the initial model because the response variable has been transformed, the overall direction and statistical significance of the major predictors remain similar. This indicates that the Box-Cox transformation primarily improves the statistical properties of the regression model rather than changing the underlying relationships between the housing characteristics and the response variable.

Table 6. Model Fit Measures for the Box-Cox Transformed Model

Model Fit Measures for the Box–Cox Transformed Model	
Measure	Value
R–squared	0.8637
Adjusted R–squared	0.8618
Residual Standard Error	3.0926
AIC	14973.6246
AICc	14974.8171
BIC	15224.9004

Table 6 summarizes several overall measures of model performance for the Box-Cox transformed regression model. The transformed model achieves an R-squared of 0.8637 and an adjusted R-squared of 0.8618, indicating that approximately 86% of the variation in the transformed response variable is explained by the selected housing characteristics. The small difference between the R-squared and adjusted R-squared values suggests that the included predictors contribute meaningfully to the model and that the model is not substantially overfitted.

In addition to the goodness-of-fit measures, the AIC, AICc, and BIC values provide useful criteria for comparing competing regression models during the model selection process. Rather than being interpreted individually, these statistics are primarily used to compare alternative models fitted to the same dataset, with smaller values generally indicating a better balance between model fit and model complexity.

Although these summary statistics suggest that the transformed model provides a satisfactory overall fit, the adequacy of the regression model also depends on whether the model assumptions are satisfied.

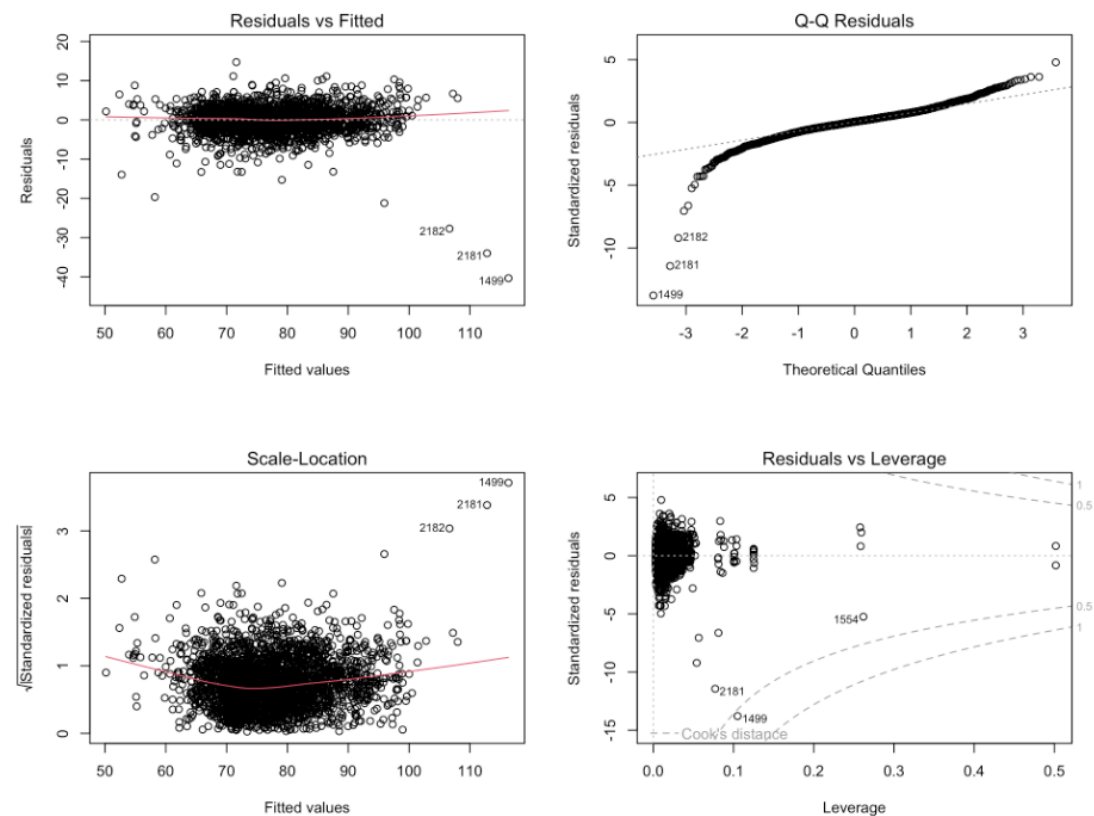


Figure 12. Standard Diagnostic Plots After the Box-Cox Transformation

Figure 12 shows the diagnostic plots for the regression model after applying the Box-Cox transformation. Compared with the initial model, the residuals appear to be more evenly scattered around zero in the Residuals versus Fitted plot. There is still a slight increase in the spread for larger fitted values, but the overall pattern looks more stable than before. This suggests that the transformation has helped reduce the unequal variance observed in the initial model.

The Q-Q plot also shows some improvement. Most of the residuals now follow the reference line reasonably well, although the observations in the lower and upper tails still deviate from the line. This indicates that the residuals are closer to a normal distribution after the transformation, but perfect normality has not been achieved.

The Scale plot supports the same conclusion. The spread of the standardized residuals is more consistent across the fitted values, although a slight upward trend remains for larger fitted values. Finally, the Residuals versus Leverage plot shows that only a few observations have relatively high leverage.

Overall, the Box-Cox transformation improves the model assumptions, especially the constant variance assumption, although some departures from normality and a few unusual observations remain.

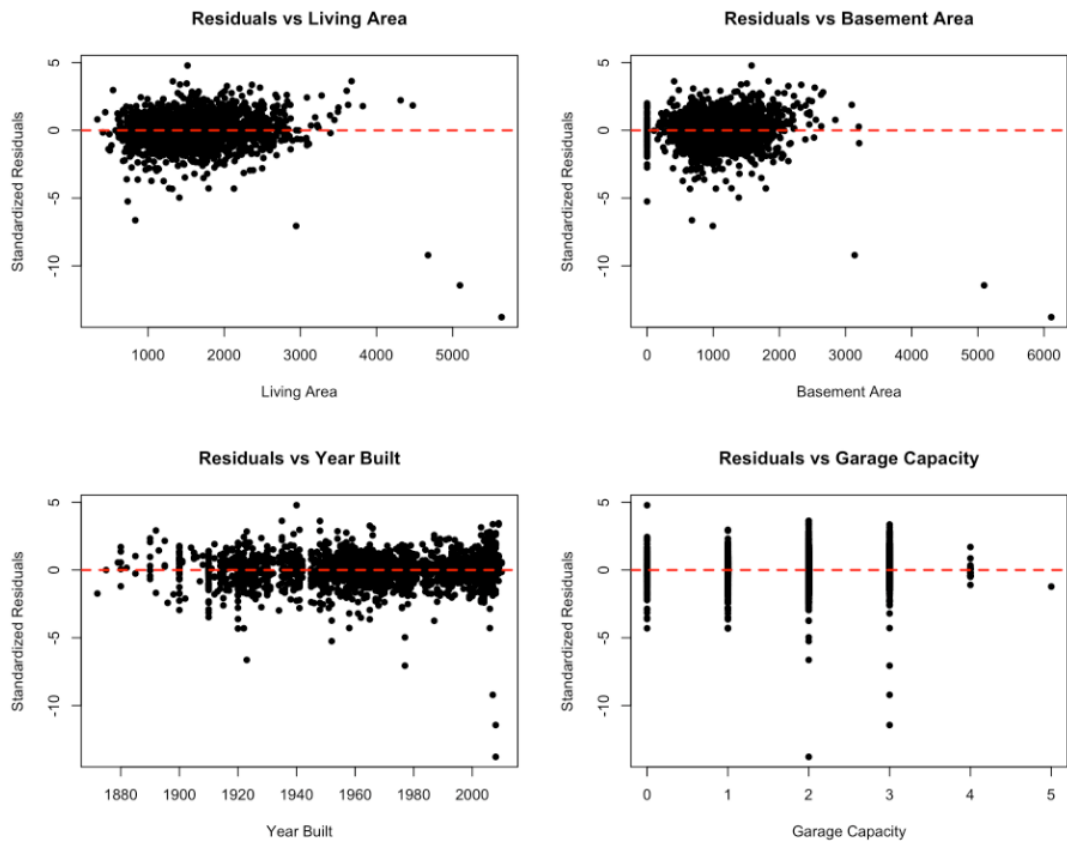


Figure 13. Residual Plots for the Numerical Predictors After the Box-Cox Transformation

Figure 13 shows the standardized residuals for each numerical predictor after applying the Box-Cox transformation. Overall, the residuals are centered around zero for all four variables, and no strong systematic patterns are observed. Compared with the initial model, the spread of the residuals is more stable across the range of each predictor, suggesting that the transformation has improved the constant variance assumption.

For living area and basement area, most observations are concentrated around zero, although a few larger houses still have relatively large residuals. This is expected

because extremely large or expensive houses are naturally more difficult to predict accurately than typical properties.

The residual plot for year built also appears more balanced than before. While a small number of recently built houses have relatively large residuals, the overall variability is more consistent across different construction years.

For garage capacity, the residuals remain centered around zero for each category.

Since garage capacity is recorded as a discrete variable rather than a continuous measurement, the vertical patterns are expected and do not indicate a problem with the model.

Overall, these residual plots suggest that the numerical predictors satisfy the regression assumptions more closely after the Box-Cox transformation, although a few unusual observations are still present.

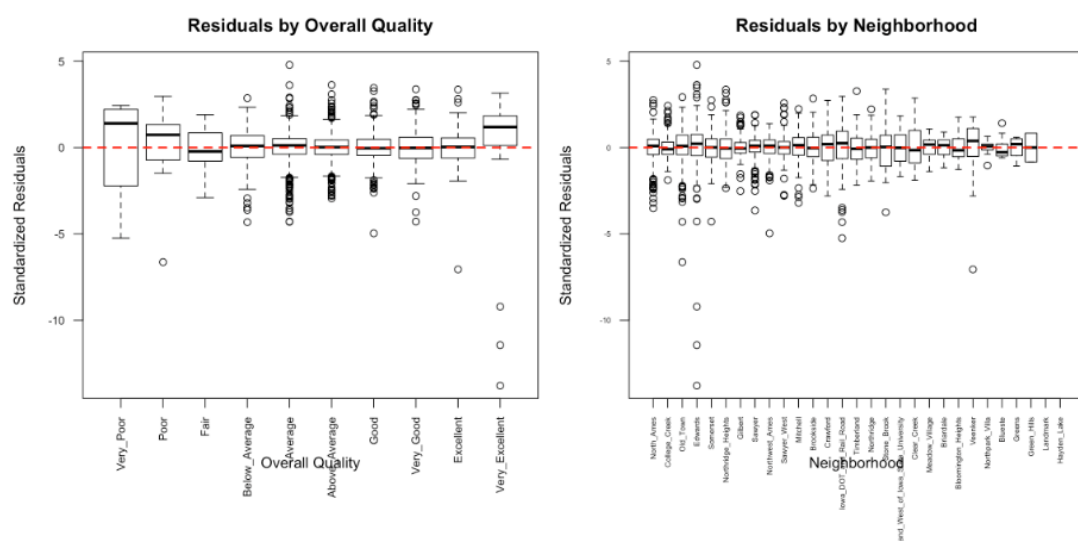


Figure 14. Residuals by Overall Quality and Residuals by Neighborhood

Figure 14 presents the standardized residuals for the two categorical predictors, overall quality and neighborhood, after applying the Box-Cox transformation.

For overall quality, the residuals for most quality levels are centered close to zero, suggesting that the model captures the average differences among quality categories reasonably well. Although several categories contain a few outlying observations, there is no obvious systematic pattern across the quality levels. This indicates that the transformation has improved the model while maintaining the relationship between housing quality and the response variable.

For neighborhood, the residuals are also generally centered around zero for most locations. Some neighborhoods show a wider spread of residuals than others, which is reasonable because housing prices within the same neighborhood can still vary depending on other housing characteristics that are not included in the model. A few neighborhoods contain several relatively large residuals, but no single neighborhood appears to consistently produce unusually large prediction errors.

Overall, these plots suggest that the transformed model fits the categorical predictors reasonably well. Although a small number of unusual observations remain, there is no strong evidence that the model systematically overestimates or underestimates housing prices for quality levels or neighborhoods.

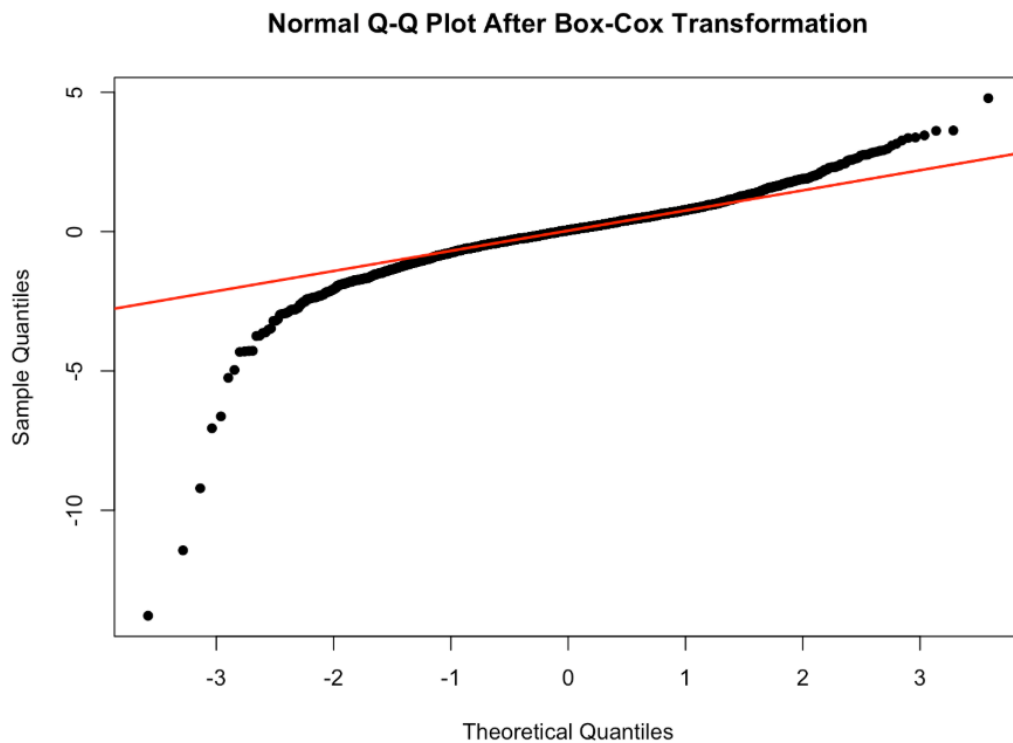


Figure 15. Normal Q-Q Plot After the Box-Cox Transformation

Figure 15 provides another assessment of the normality assumption after applying the Box-Cox transformation. Compared with the Q-Q plot from the initial regression model, the residuals follow the reference line more closely over most of the distribution. In particular, the points in the middle of the plot are almost aligned with the reference line, suggesting that the transformation has improved the normality of the residuals.

Some departures from the reference line are still visible in both tails, especially in the lower tail where a few observations fall noticeably below the line.

Overall, the Q-Q plot suggests that the Box-Cox transformation improves the normality assumption compared with the initial regression model, although perfect

normality is still not achieved. Since the residuals now show a more reasonable pattern and the other diagnostic plots also indicate improved model performance, the transformed model provides a better basis for the remaining model selection procedures.

Identify Multicollinearity

Before selecting the final regression model, multicollinearity was examined using the generalized variance inflation factor (GVIF). Because overall quality and neighborhood are categorical variables with multiple levels, the adjusted GVIF values were used for interpretation instead of the raw GVIF values.

Table 7. Generalized Variance Inflation Factors (GVIF)

Generalized Variance Inflation Factors for the Box–Cox Transformed Model			
Variable	GVIF	Degrees of Freedom	Adjusted GVIF
living_area	1.986	1	1.409
basement_area	1.833	1	1.354
overall_quality	6.749	9	1.112
year_built	5.190	1	2.278
neighborhood	18.244	27	1.055
garage_capacity	2.011	1	1.418

Before selecting the final regression model, multicollinearity was assessed using the generalized variance inflation factor (GVIF). Since overall quality and neighborhood are categorical variables with multiple levels, the adjusted GVIF values were used to evaluate multicollinearity.

As shown in Table 7, all adjusted GVIF values are relatively small, ranging from 1.055 to 2.278. The largest adjusted GVIF is observed for year built, with a value of 2.278, while the remaining predictors all have adjusted GVIF values close to 1. These

results indicate that the explanatory variables are not highly correlated with one another.

Based on these results, there is no evidence of serious multicollinearity in the transformed regression model. Therefore, all selected predictors were retained for the subsequent model selection process. The next step is to investigate whether adding a quadratic term for year built can further improve the model fit.

Table 8. Partial F-test Comparing the Transformed Model and the Quadratic Model

Partial F–test Comparing the Transformed Model and the Quadratic Model						
Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)	
2889	27630.18	NA	NA	NA	NA	
2888	27629.66	1	0.5135	0.0537	0.8168	

To investigate whether a nonlinear relationship exists between year built and the transformed response variable, a quadratic term for year built was added to the regression model. A partial F-test was then used to compare the transformed model with the model including the quadratic term.

As shown in Table 8, adding the quadratic term produces only a very small reduction in the residual sum of squares. The partial F-test gives an F-statistic of 0.0537 with a p-value of 0.8168, indicating that the quadratic term does not provide a statistically significant improvement in model fit. In addition, the coefficient of the quadratic term

is also not statistically significant ($p = 0.8168$), suggesting that there is no evidence of a nonlinear relationship between year built and the transformed response variable.

Based on these results, the quadratic term was not included in the final regression model. The simpler transformed model was retained because it provides a similar fit while avoiding unnecessary model complexity.

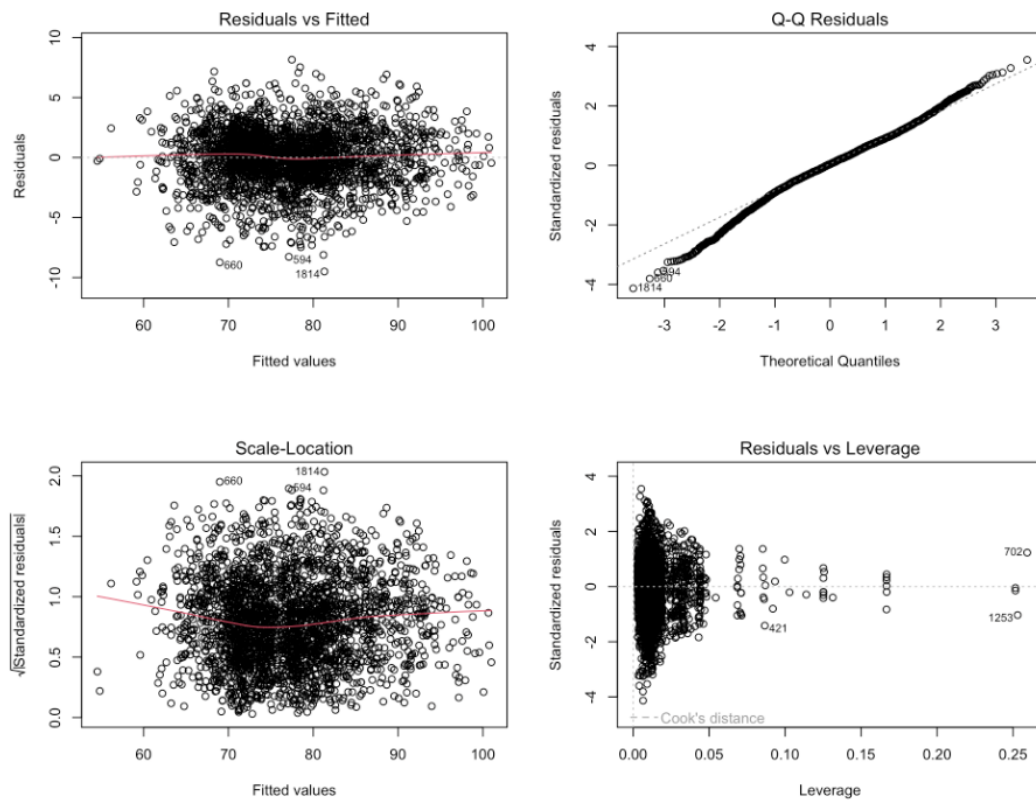


Figure 16: Diagnostic plots for the transformed regression model after Box-Cox transformation

Figure 16 is showing the diagnostic plots of the transformed regression model.

Although slight deviations from normality remain in the tails of the Q-Q plot, most of

the residuals lie close to the reference line, suggesting that the normality assumption has improved after the Box-Cox transformation. A small number of observations exhibit relatively high leverage. Therefore, Cook's distance was used to identify influential observations before fitting the final model.

Table 9 Summary of Cook's Distance Results

Summary of Cook’s Distance Results	
Measure	Value
Total observations	2.930e+03
Cook’s distance threshold	1.365e−03
Observations above threshold	1.620e+02

Table 10 Comparison Before and After Removing Influential Observations

Comparison Before and After Removing Influential Observations					
Model	Observations	R_squared	Adjusted_R_squared	Residual_Standard_Error	AIC BIC
Transformed Model	2930	0.8637	0.8618	3.0926	14973.62 15224.90
Model After Removing Influential Observations	2768	0.9085	0.9072	2.3051	12519.31 12756.34

Table 10 shows that 162 observations exceeded the Cook's distance threshold of $4/n$ (approximately 0.00137), indicating that these observations had relatively large influence on the fitted regression model and were removed before fitting the final model. And as shown in Table10, removing influential observations improved the overall model performance. The R^2 increased from 0.8637 to 0.9085, while the adjusted R^2 increased from 0.8618 to 0.9072. The residual standard error decreased from 3.0926 to 2.3051, and both AIC and BIC were substantially reduced, indicating a

better fitting and more parsimonious model.

Table 11: Coefficient Estimates for the Final Regression Model

Coefficient Estimates for the Final Regression Model						
Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-37.0949	6.8528	-5.4131	0.0000	-50.5321	-23.6576
living_area	0.0054	0.0001	41.9165	0.0000	0.0052	0.0057
basement_area	0.0028	0.0001	19.2802	0.0000	0.0025	0.0030
overall_qualityFair	5.2070	1.2315	4.2283	0.0000	2.7923	7.6218
overall_qualityBelow_Average	7.7802	1.1724	6.6362	0.0000	5.4814	10.0791
overall_qualityAverage	9.7108	1.1658	8.3301	0.0000	7.4250	11.9967
overall_qualityAbove_Average	10.9388	1.1684	9.3620	0.0000	8.6477	13.2299
overall_qualityGood	12.3350	1.1743	10.5042	0.0000	10.0324	14.6376
overall_qualityVery_Good	14.6577	1.1875	12.3428	0.0000	12.3291	16.9863
overall_qualityExcellent	18.7491	1.2173	15.4027	0.0000	16.3623	21.1360
overall_qualityVery_Excellent	19.8398	1.3713	14.4682	0.0000	17.1510	22.5286
year_built	0.0456	0.0034	13.2731	0.0000	0.0388	0.0523
neighborhoodCollege_Creek	0.5375	0.2241	2.3985	0.0165	0.0981	0.9770
neighborhoodOld_Town	-1.1798	0.2288	-5.1552	0.0000	-1.6285	-0.7310
neighborhoodEdwards	-0.8697	0.2128	-4.0875	0.0000	-1.2869	-0.4525
neighborhoodSomerset	0.8948	0.2659	3.3648	0.0008	0.3734	1.4163
neighborhoodNorthridge_Heights	1.9392	0.3023	6.4144	0.0000	1.3464	2.5321
neighborhoodGilbert	0.2323	0.2578	0.9013	0.3675	-0.2731	0.7378
neighborhoodSawyer	0.0693	0.2212	0.3134	0.7540	-0.3645	0.5031
neighborhoodNorthwest_Ames	0.1361	0.2447	0.5562	0.5781	-0.3436	0.6158
neighborhoodSawyer_West	-0.2737	0.2640	-1.0367	0.2999	-0.7914	0.2440
neighborhoodMitchell	0.2844	0.2584	1.1006	0.2712	-0.2223	0.7910
neighborhoodBrookside	-0.0095	0.2697	-0.0351	0.9720	-0.5383	0.5193
neighborhoodCrawford	3.5470	0.2766	12.8224	0.0000	3.0046	4.0894
neighborhoodIowa_DOT_and_Rail_Road	-1.5139	0.3039	-4.9812	0.0000	-2.1099	-0.9180
neighborhoodTimberland	1.5526	0.3358	4.6230	0.0000	0.8941	2.2111
neighborhoodNorthridge	2.2122	0.3617	6.1168	0.0000	1.5030	2.9213
neighborhoodStone_Brook	2.2561	0.4621	4.8826	0.0000	1.3501	3.1621
neighborhoodSouth_and_West_of_Iowa_State_University	-1.2844	0.3865	-3.3234	0.0009	-2.0423	-0.5266
neighborhoodClear_Creek	2.4040	0.4153	5.7885	0.0000	1.5896	3.2183
neighborhoodMeadow_Village	-3.4694	0.4182	-8.2953	0.0000	-4.2895	-2.6493
neighborhoodBriardale	-4.1951	0.4464	-9.3976	0.0000	-5.0704	-3.3198

Table 11 reports the coefficient estimates for the final regression model after removing influential observations. Most explanatory variables remain statistically significant at the 5% significance level. Living area, basement area, year built, garage capacity, overall quality, and several neighborhood indicators continue to show significant associations with the transformed response variable. A few neighborhood categories, such as Gilbert, Sawyer, Northwest Ames, Mitchell, Brookside, Greens, and Landmark, are not statistically significant after accounting for the other housing characteristics. \$Because the response variable was transformed using the Box-Cox

transformation, the regression coefficients should not be interpreted as direct dollar changes in housing sale price. Instead, the coefficients describe the direction and relative magnitude of the association between each predictor and the transformed response variable while holding all other variables constant.

Table 12: Model Fit Measures for the Final Regression Model

Model Fit Measures for the Final Regression Model	
Measure	Value
Number of Observations	2768.0000
R-squared	0.9085
Adjusted R-squared	0.9072
Residual Standard Error	2.3051
F Statistic	712.6644
Model p-value	0.0000
AIC	12519.3073
BIC	12756.3425

Table 12 summarizes the overall performance of the final regression model.

Compared with the transformed model before removing influential observations, the final model shows a noticeable improvement in model fit. The R^2 and adjusted R^2 both increase, while the residual standard error, AIC, and BIC all decrease, indicating a better balance between goodness of fit and model complexity. Overall, the final model explains approximately 91% of the variation in the transformed response

variable and provides the best fit among all models considered in this study.

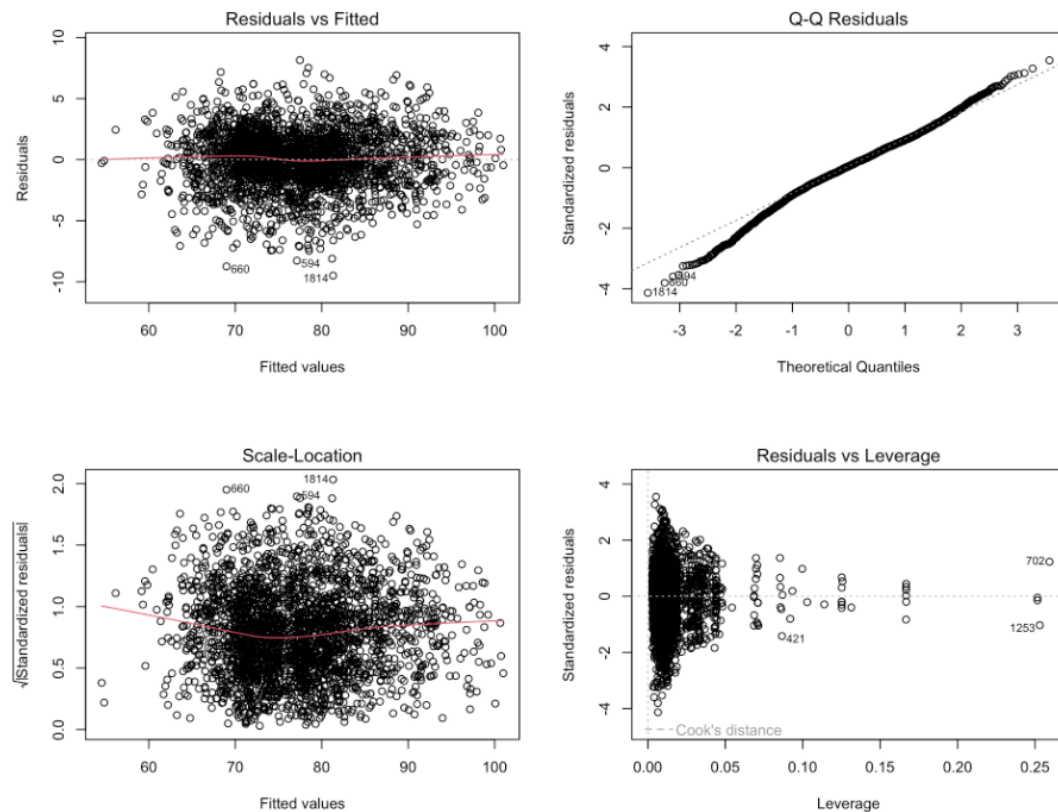


Figure 17: Diagnostic plots for the final regression model

Figure 17 presents the diagnostic plots for the final regression model. Compared with the previous model, the residuals are more evenly scattered around zero, and the Scale-Location plot indicates a more stable variance across the range of fitted values. The Q-Q plot shows that most residuals follow the reference line reasonably well, although slight departures remain in the tails. The Residuals versus Leverage plot suggests that no observations have an excessively large influence on the final fitted model. Overall, the regression assumptions are reasonably satisfied.

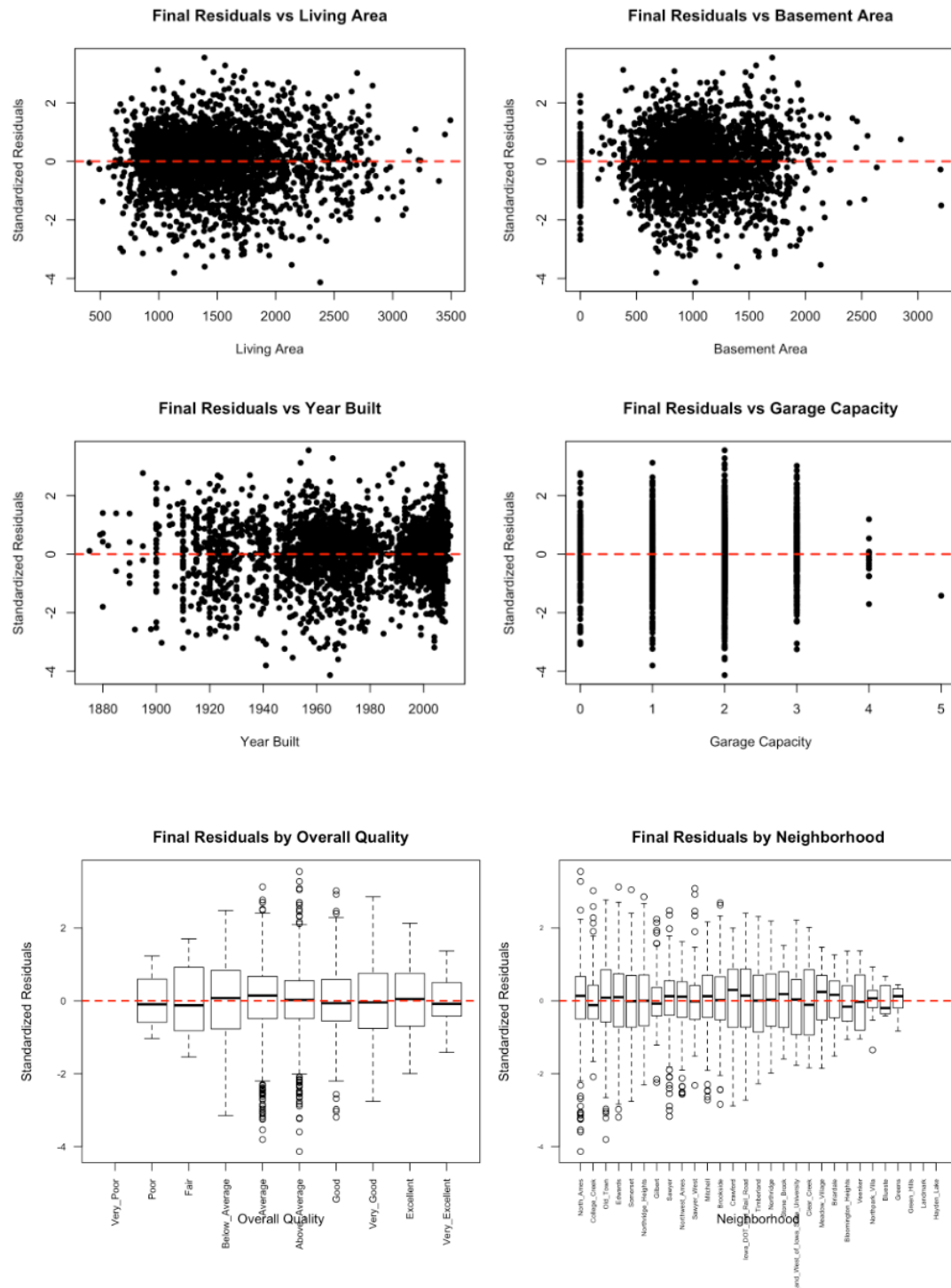


Figure 18: Residual plots by predictors

Figure 18 displays the standardized residuals against each explanatory variable. For the numerical predictors, the residuals are randomly distributed around zero without

obvious systematic patterns, suggesting that the linearity assumption is appropriate.

The residuals also show relatively constant variability across the ranges of living area, basement area, year built, and garage capacity.

For the categorical predictors, the residuals are generally centered around zero within each level of overall quality and neighborhood. Although several categories contain a small number of outlying observations, no strong systematic pattern is observed.

These diagnostic plots further support the adequacy of the final regression model.

Overall, after applying the Box-Cox transformation and removing influential observations identified by Cook's distance, the final regression model achieved the best overall performance. The diagnostic plots indicate that the model assumptions are reasonably satisfied, while the model fit statistics demonstrate substantial improvement over the initial and transformed models. Therefore, I choose that model as my final model, and I will discuss this model interpret.

Model Validation

To evaluate the model's predictive performance, the dataset was randomly split into a training set and a testing set. The model was fitted using the training set, and predictions were generated for the testing set. Because the response variable was log-transformed during model fitting, predictive performance was first evaluated on the log scale using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). In addition, the predicted values were transformed back to the original sale price scale to calculate RMSE, MAE, and R^2 , providing a more interpretable assessment of prediction accuracy in terms of actual housing prices. The results indicate that the model achieved reasonably good predictive performance on the testing set.

Table 13. Sample Sizes for the Training and Test Sets

Sample Sizes for the Training and Test Sets	
Dataset	Observations
Training Set	2214
Test Set	554

Table 13 shows that the training dataset contains 2,214 observations, while the remaining 554 observations form the independent test dataset.

Table 14. Test-Set Performance of the Final Regression Model

Test-Set Performance
of the Final Regression
Model

Measure	Value
Test MSE	5.5039
Test RMSE	2.3460
Test MAE	1.7810
Test R-squared	0.8978

Table 14 summarizes the predictive performance of the final regression model on the independent test dataset. The model achieved a test R^2 of 0.8978, indicating that approximately 89.8% of the variation in the transformed response variable can be explained for unseen observations. The test RMSE (2.3460) is close to the residual standard error (2.3051) obtained from the final model, suggesting that the model generalizes well without substantial overfitting.

Table 15. Test-Set Performance on the Original Sale Price Scale

Test–Set Performance on the Original Sale Price Scale

Measure	Value
Test RMSE in Dollars	20488.94
Test MAE in Dollars	15140.63
Test R–squared	0.90

Table 15 reports prediction accuracy after converting the Box–Cox transformed predictions back to the original sale price scale. The model achieved an RMSE of approximately \$20,489 and an MAE of approximately \$15,141 while maintaining an R^2 of 0.90. These results indicate that the model provides accurate predictions of housing sale prices on the original measurement scale.

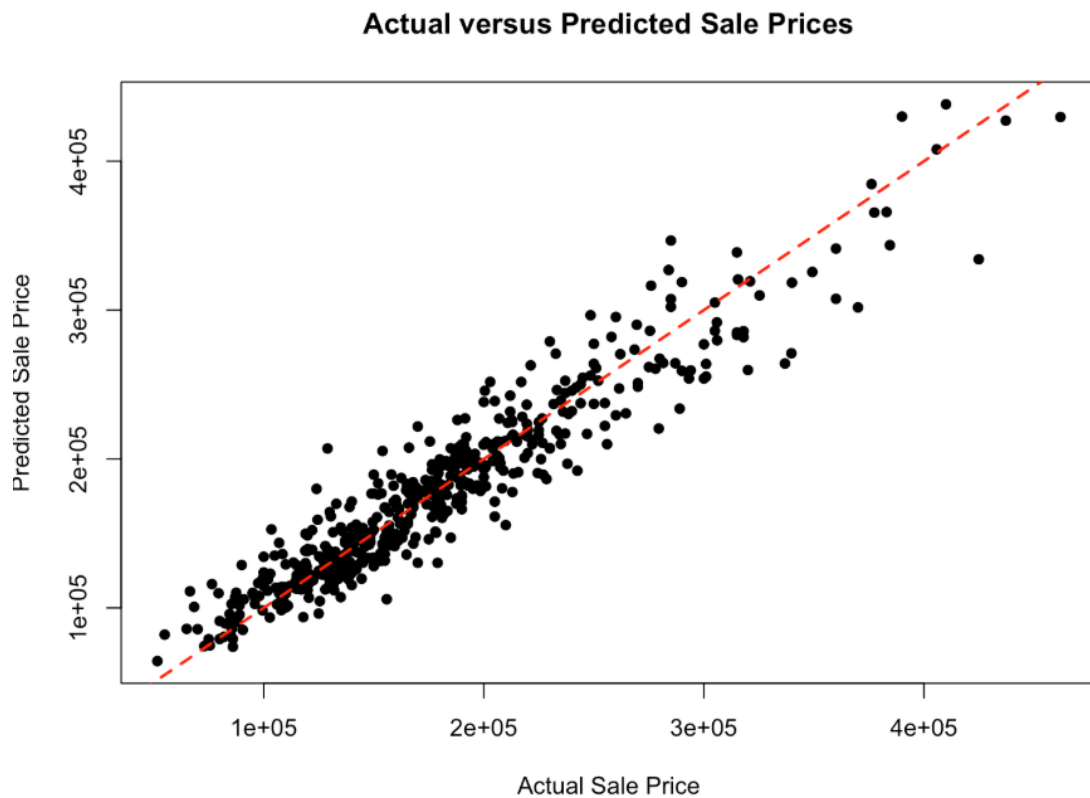


Figure 19. Actual versus Predicted Sale Prices

Figure 19 compares the predicted sale prices with the observed sale prices in the test dataset. Most observations lie close to the 45-degree reference line, indicating strong agreement between predicted and actual values. Although prediction errors become slightly larger for a small number of higher-priced homes, no substantial systematic bias is observed. Overall, Figure 19 shows that the final regression model predicts housing sale prices accurately across a wide range of values.

Together with the diagnostic analyses presented in Section 7, these validation results provide strong evidence that the final regression model is both statistically appropriate and practically useful for explaining and predicting housing sale prices in the Ames

Housing dataset.

Conclusion

This study investigated the relationship between housing characteristics and residential sale prices using the Ames Housing dataset. Multiple linear regression was employed to examine the effects of living area, basement area, overall quality, year built, neighborhood, and garage capacity on housing prices. To improve the validity of the regression assumptions, a Box–Cox transformation was applied to the response variable, and influential observations identified using Cook’s distance were removed before fitting the final model.

The final regression model demonstrated strong explanatory and predictive performance. Living area, basement area, overall quality, year built, and garage capacity were all statistically significant predictors of the transformed sale price. In addition, several neighborhood indicators remained significant after controlling for the other housing characteristics, suggesting that location also contributes to housing value. After removing influential observations, the model achieved an R^2 of 0.9085 with an adjusted R^2 of 0.9072, while the residual standard error decreased substantially compared with the earlier model.

Model validation further confirmed the robustness of the final model. The test-set evaluation produced a test R^2 of approximately 0.90, and the prediction errors on the original sale price scale remained reasonably small, with an RMSE of approximately \$20,489 and an MAE of approximately \$15,141. The close agreement between the training and test performance measures suggests that the model generalizes well to

unseen observations without substantial overfitting.

Although the final model performs well, several limitations should be acknowledged.

The analysis considers only linear relationships between the predictors and the transformed response variable, and interaction effects among predictors were not explored. In addition, the results are based on the Ames Housing dataset and may not generalize directly to other housing markets. Future research could investigate nonlinear regression models, interaction terms, or machine learning methods to further improve prediction accuracy.

Overall, the final regression model provides an accurate and interpretable framework for explaining and predicting housing sale prices. The results demonstrate that multiple housing characteristics jointly influence sale prices and that careful model selection, diagnostic checking, transformation, and validation can substantially improve regression performance.